

Securing Shared University Lab Computers

A Practical, Free, Administrator-Led Approach to Stopping Credential Leakage on Shared Endpoints

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Background

Every university runs shared computer laboratories, and almost all of them run those machines in the default configuration that Windows and the Chrome or Edge browser ship with. That default configuration was designed for a personal computer owned by one person — not for a machine that dozens of different students use in a single day. The result is a quiet but serious security gap. On a typical shared lab computer, one student's saved passwords, synchronised personal data, and active sessions can be recovered by the next person who sits down at the same machine. No malware is installed, no software flaw is exploited, and no password is cracked. The system simply behaves the way it was built to behave, and in a shared setting that behaviour leaks private information from one user to the next.

This is not a theoretical concern. Credential theft of exactly this kind has led to real harm in Bangladesh, including cases where a student's personal accounts were taken over and private images were leaked, and cases where examination materials were stolen after staff credentials were captured on a shared machine. The people most exposed are often those least able to defend themselves — students with little technical background, and especially women, for whom the consequences of a compromised account can be severe. Despite this, no documented, low-cost, reproducible defence for shared lab endpoints exists for universities in Bangladesh, and the problem is largely left unaddressed.

How the Leakage Happens

The exposure comes from a small number of distinct weaknesses, each rooted in normal, intended behaviour. The first is profile persistence. When a student signs into the browser, saves passwords, and later leaves without fully logging out, the browser profile remains on the machine. The next user can open the browser's password settings, enter the shared local password, and read or export every stored credential in seconds, without any special tools.

The second weakness is synchronisation. When a student enables browser sync — something the browser actively encourages — their entire personal browsing world is downloaded onto the lab machine: passwords, autofill data, browsing history, and in some cases photographs and other data from their personal devices. All of this is written to the lab computer's disk, where it remains until something removes it.

The third weakness is residual data on disk. Even after a user signs out, the profile folder and its files remain. Freely available recovery utilities can read the stored credentials directly from these files, again requiring no administrative rights. The fourth weakness is the keylogger: a malicious student can install a small hidden program, or a malicious browser extension, that records every keystroke typed by everyone who uses the machine afterwards — capturing passwords, private messages, and anything else entered. The fifth and hardest weakness is the hardware keylogger, a small physical device placed between the keyboard and the computer that records all input and cannot be detected by any software.

The Guiding Principle

The research literature on security is consistent on one point: training users to be more careful produces, at best, small and short-lived improvements. People are busy, the warnings are easy to ignore, and a single

lapse is enough. The approach taken here is therefore different. Rather than asking users to change their behaviour, the goal is to change the system so that the opportunity for harm is removed before it can arise. If the dangerous data is never written to the machine, and is destroyed the moment a session ends, then there is nothing for the next person to find. Every measure proposed below is an administrator-side change to the machine itself, not an instruction given to the user, and every one of them uses software that is either already built into Windows and the browser or is free and open-source. The total additional software cost is zero.

The Proposed Solution

The defence rests on a small backbone of four measures that together close most of the attack surface. The first is to ensure that students use the lab machines as standard users rather than administrators, which removes their ability to install keyloggers or disable protections. The second is to give each machine a reliable reset mechanism — either Windows Shared PC Mode, which deletes a user’s profile the moment they log out, or the Unified Write Filter, which returns the machine to a clean state on every reboot. Either one ensures that nothing a user leaves behind survives into the next session.

The third measure is to lock down the browser through centrally managed policy: disabling sign-in and synchronisation so personal data never arrives, disabling the built-in password manager so nothing is stored, forcing fresh temporary profiles, and blocking the installation of unapproved extensions so malicious add-ons cannot take hold. The fourth measure is detection. Because no prevention is perfect, the free monitoring tools Sysmon and Wazuh are deployed to watch for the tell-tale signs of an attack in progress — an attempt to copy the credential database, a hidden script capturing keystrokes, or data being sent to an unusual destination — and to alert administrators when prevention has been bypassed.

Around this backbone sit several supporting measures. Application control through AppLocker, together with a restricted PowerShell mode, prevents unapproved programs and scripts — including the typical homemade keylogger — from running at all. A Defender rule guards against tools that try to steal credentials directly from system memory. For the hardware keylogger, which software genuinely cannot defeat, the honest answer is physical: controlling which devices may connect, inspecting machines, and using the on-screen keyboard for the most sensitive logins so that there are no physical keystrokes for such a device to capture. This limitation is stated plainly rather than glossed over, because an honest account of what the defence cannot do is part of what makes the rest of it trustworthy.

Demonstrating That It Works

A proposal of this kind is only convincing if its effect can be measured. The plan is therefore to apply the measures to one laboratory while leaving a second, comparable laboratory unchanged as a point of comparison. Before any change is made, a controlled test — using only accounts created by the research team, never the data of real students — establishes how many machines in each laboratory currently leak credentials. The measures are then deployed to the first laboratory over a thirty-day period, after which the same test is repeated. The expectation is that the treated laboratory will move from leaking on most machines to leaking on none, while the untouched laboratory remains exactly as it was. An anonymous survey of students, conducted before and after, captures the human side: how aware they are of the risk, how they actually behave at the machines, and whether that awareness shifts. The headline outcome is a single clear number — the change in how many machines expose credentials — supported by the survey and by the monitoring records.

What Will Be Built

Three small tools support the work and form its original technical contribution. The first is an audit script that examines a machine and reports exactly which protections are present and which are missing; it

changes nothing and exists only to measure. The second is a disposable-profile launcher that opens the browser inside a temporary folder with synchronisation forced off, and deletes that folder completely when the browser closes, leaving no trace of the session — with the honest scope that it addresses the synchronisation and profile-residue problem and does not, by itself, stop keyloggers. The third is a test harness that seeds a machine with a controlled test account, attempts to recover the stored credentials, and records whether it succeeded, providing the before-and-after measurement that proves the effect. Together with the policy configuration, these tools make the whole approach reproducible: any other institution facing the same problem can take them and apply the same defence.

In short, the shared lab computer leaks one user's private data to the next because of how it is configured by default. The fix is free, built from tools already available, deployable by an administrator without buying anything, reversible if it causes trouble, and measurable in its effect. It will not stop a physical hardware keylogger, and it does not pretend to. Everything else, it closes.

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